CS550/DSL501: Machine Learning (2024–25–M)

CauESC: A Causal Aware Model for Emotional Support Conversation

Project Report(STRIKERS)

1 Introduction

The primary objective of this project is to develop an Emotional Support Conversation (ESC) model based on causal awareness techniques. Our approach utilizes the framework proposed in "CauESC: A Causal Aware Model for Emotional Support Conversation," which emphasizes recognizing the causes behind emotional distress and generating supportive responses accordingly. This report outlines the work completed so far, including dataset curation, exploration of transformer models, and the initial implementation phase of our cause-aware model.

2 Problem Formulation

The task of ESC can be mathematically formulated as follows:

Input Components:

- Situation (q): Describes the seeker's problem in free-form text.
- Conversation History (C): A sequence of utterances $C = (u_1, u_2, \ldots, u_M)$.
- Strategy History (S): Strategies employed by the supporter $S = (s_1, s_2, \ldots, s_J)$.

Objective: To estimate the probability distribution for generating a supportive response y:

$$p(y \mid q, C, S)$$

This formulation highlights the need for a model that can effectively integrate these components to produce meaningful and supportive dialogue.

3 Approach Overview

The development of our CauESC model for the machine learning project consists of three main layers:

- Cause Aware Encoder
- Causal Interaction Module
- Independent-Integrated Strategy Executors

4 Dataset Curation

To train the ESC model, we have curated a dataset from multiple sources, ensuring it contains a variety of emotionally supportive conversations with annotations that include the causes of emotional distress and different strategies used for emotional support. The dataset selection process involved:

- **Selection and annotation of dataset:** We selected publicly available datasets such as **ESConv**, which is well-known for emotional support conversations. Additionally, we sourced dialogues from forums and platforms where supportive conversations occur, ensuring diversity in dialogue styles and emotional contexts. Then we annotated the dataset to mark emotion causes, strategies employed, and emotional effects. This annotation included identifying the emotional triggers in each dialogue and categorizing the type of support strategy used (e.g., reflection of feelings, providing suggestions).
- **Data Pre-processing:** We filtered the dataset to remove any sensitive or inappropriate content, ensuring it aligns with ethical guidelines for training models designed to interact with emotionally vulnerable users.

5 Exploration and development of Transformer Models

We have explored various transformer-based models to understand their capabilities and potential use cases in our project. Key focus areas included:

- **COMET** (Commonsense Transformers) for Causal Reasoning:
  - **Functionality Exploration:**COMET, trained on the ATOMIC knowledge base, generates commonsense inferences about events and their associated emotional responses. It can infer intra and inter effects based on given scenarios, making it suitable for reasoning about emotional causes and effects.

- **Adaptation for Our Use Case:** Instead of using the traditional context `C` as the input to COMET, we opted to feed the **supporter's utterances** as input. This approach helps generate the anticipated emotional effects based on the supporter's contributions to the conversation, allowing us to capture the causal dynamics within the dialogue.
- **Implementation Strategy:** We implemented input substitution to use the supporter's utterances as events for COMET's causal reasoning. The output from COMET will be used as part of the input to our cause-aware encoder to enhance the model's understanding of the emotional dynamics.

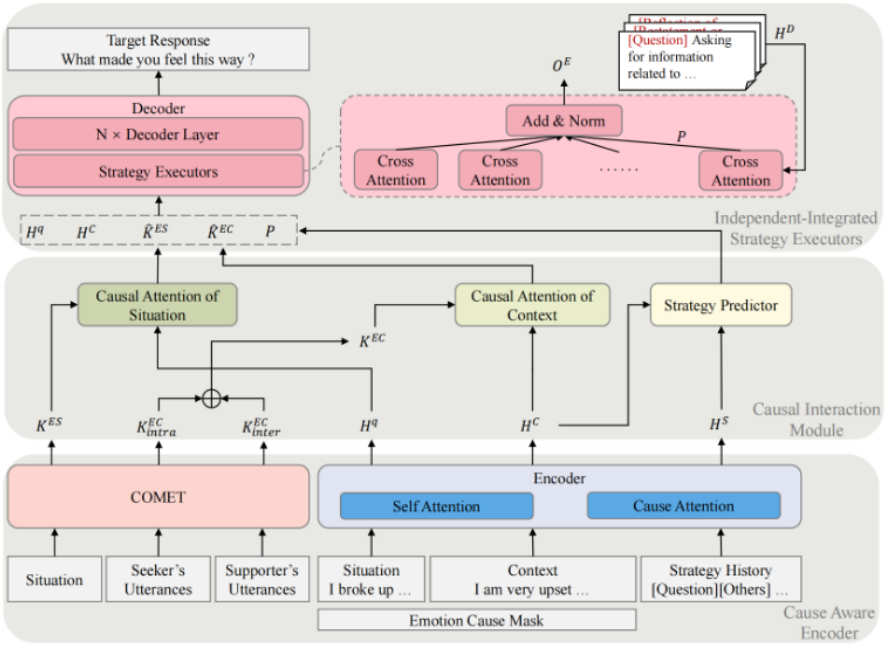
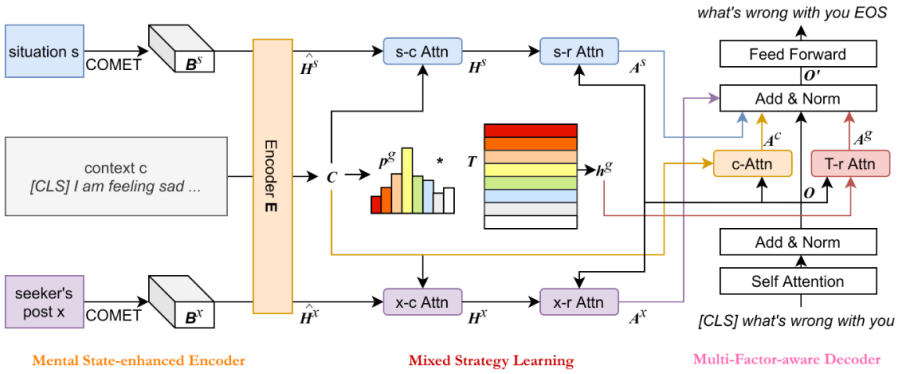


Figure 1: Enter Caption

Development of the First Layer of the Cause-Aware Model: The first layer of our cause-aware model has been designed to recognize and encode the causes of emotional distress. The steps involved in this phase included:

- - **Cause Recognition:** We utilized a pre-trained emotion cause detector (fine-tuned on datasets like RECCON) to identify emotion-triggering utterances within a given conversation.

- **Cause-Aware Encoding:** Implemented a custom attention mechanism (`Cause Attention`) to prioritize the utterances identified as causes of distress. This attention layer is designed to enhance the representation of the emotional causes in the dialogue context, improving the model's ability to generate supportive responses.
- **Integration with COMET Outputs:** The outputs from COMET's causal reasoning (using the supporter's utterances) were fed into the encoder to provide a richer understanding of the emotional landscape, including the effects of specific support strategies.



6 Challenges faced

- **Data Annotation Consistency:** Ensuring consistent labeling across different datasets required manual verification and adjustments.
- **Fine-tuning COMET for Specific Use Cases:** Adapting COMET's outputs to align with the emotion recognition and cause-effect reasoning required careful configuration and experimentation.

7 Theoretical Framework and Formulas

7.1 Cause Aware Encoder

The Cause Aware Encoder is designed to identify the emotional causes of the seeker's distress and capture the emotional effects within the conversation. This component can be further divided into two tasks:

7.1.1 Emotion Cause Encoding

To identify emotional causes, an existing emotion cause detector is utilized, trained on an open-domain emotional dialogue dataset called RECCON. The

detector determines which utterances in a context cause emotions in the target utterance.

Input: Context, target utterance, and emotion label.

Output: Identification of sentences causing the seeker's emotion.

The Cause Attention mechanism emphasizes utterances with emotional causes. The cause attention distribution a_{ij} is computed as follows:

$$a_{ij} = \frac{e_i \cdot \exp\left(\frac{q_i k_j}{d}\right)}{\sum_{n=1}^{N} e_n \cdot \exp\left(\frac{q_i k_n}{d}\right)}$$

Where:

- q_i: Query vector for the ith position.
- k_j: Key vector for the jth position.
- d: Dimension of the model.

The outputs from both self-attention and cause attention are fused using:

$$\mu = \text{ReLU}(W_\mu[A_c; A_s] + b_\mu)$$

$$F = \mu \cdot A_s + (1 - \mu) \cdot A_c$$

Where:

- W_μ, b_μ: Learnable parameters.
- A_c, A_s: Outputs from cause and self-attention.

7.1.2 Emotion Effect Acquisition

To obtain emotion effects based on emotional dynamics, COMET (a commonsense reasoning model) is employed. It categorizes relationships into intra-relations (e.g., xReact) and inter-relations (e.g., oReact). The input to COMET is structured as follows:

$$\text{input} = (u_i[\text{MASK}]o\text{React})$$

Where u_i is an event or utterance.

The overall emotion effects are captured through:

$$K_{ES} = \text{COMET}(rel, s)$$

Where rel represents various relation types based on causal relations.

7.2 Causal Interaction Module

This module reasons about the emotion effects triggered by emotion causes to develop a fine-grained understanding of emotions during interactions.

To consider emotion effects comprehensively, we concatenate the previously obtained effects:

$$\hat{K}_{EC} = \text{CROSS-ATT}(K_{EC}, H_C)$$

$$\hat{K}_{ES} = \text{CROSS-ATT}(K_{ES}, H_q)$$

Where CROSS-ATT denotes a combination of cross-attention mechanisms with residual connections.

The strategy history is integrated with the context representation through mean-pooling:

$$s = \text{Mean-Pooling}(H_S), \quad c = \text{Mean-Pooling}(H_C)$$

Finally, we estimate the strategy distribution for strategy modeling using:

$$p_i = \frac{e^{h^T k_i}}{\sum_{j=1}^{n} e^{h^T k_j}}$$

Where:

- h: Query vector formed from mean pooling.
- k_i: Key vector for strategy i.

7.3 Independent-Integrated Strategy Executors

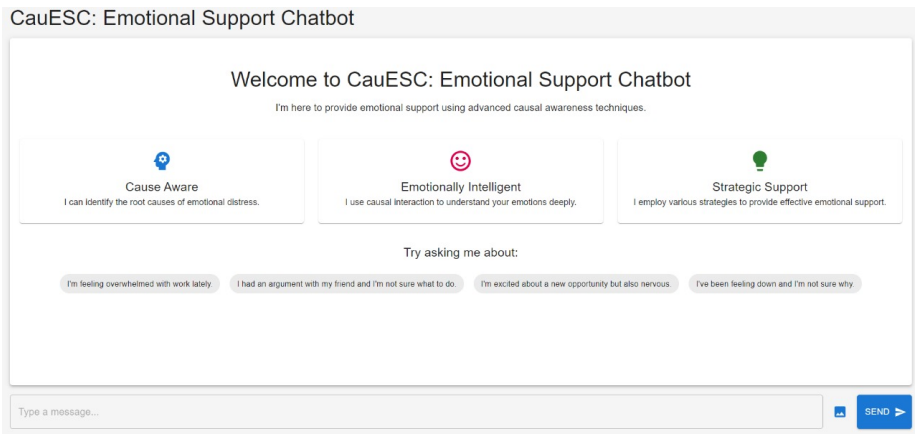
Existing models often assume that systems implicitly learn how to respond supportively without additional inductive bias. CauESC introduces independent-integrated strategy modeling to enhance interpretability and effectiveness in response generation.

This approach allows for both independent comprehension of strategies and their integrated application during response generation.

8 Next Steps

- **Implementation of Causal Interaction Module:** The next step involves developing the causal interaction module, which will reason about the effects triggered by the emotional causes using the encoded cause-aware context and COMET outputs.
- **Strategy Modeling:** Work on integrating the independent-integrated strategy modeling, where different support strategies will be used to generate the response.

- **Evaluation:** We will start evaluating the model's performance using automatic metrics (e.g., BLEU, ROUGE) and human evaluations to refine the model further.
- We can define additional layers for the MISCModel class and can also define mental state and order where we would typically use COMET to get mental state information.



9 Contributions

Implementing the CauESC conversation model for this machine learning project includes contributions from the team members as follows:

- Akshat Kumar(12240080): Developed the code developed till now and theory of the project.
- Aditya Prakash(12240040): Developed the code developed till now and theory of the project.
- Anand Vardhan(12240140): Formulated the project report and theoretical buildup of the project.
- Shashank Pant(12241700): Theoritical buildup and developing the User Interface of the chat system for CauESC Model.

The project is progressing as planned, with the completion of dataset curation and the initial implementation of the cause-aware model's first layer. The integration of COMET for causal reasoning shows promise in enhancing the model's understanding of emotional effects. The upcoming phases will focus on further refining the model's architecture and evaluating its performance in generating supportive responses.

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